29_DATABASE_SCHEMA.md

UNPLND Database Schema

Version 1.0

Priority: CRITICAL

Status: MVP

Purpose

This document defines the complete Firestore database structure.

The schema must remain scalable, secure and easy to maintain.

Every collection has a clear responsibility.

Avoid duplicated data whenever possible.

Database Principles

Normalize where practical.

Denormalize only when it improves performance significantly.

Never duplicate user-generated content unnecessarily.

Use document references when appropriate.

Root Collections

/users

/plans

/albums

/favorites

/collections

/discover

/notifications

/app_config

/feedback

Users Collection

Collection

/users/{userId}

Fields

userId

displayName

email

photoURL

premium

createdAt

updatedAt

language

theme

locationPermission

notificationPermission

favoriteMoods

favoriteCategories

stats

settings

The user document should remain lightweight.

User Statistics

Stored inside

users/{userId}/stats

Examples

plansCreated

plansCompleted

albumsCreated

albumsShared

citiesVisited

favoriteMood

favoriteBudget

totalDistance

totalWalkTime

Memory statistics should be precomputed where possible.

Plans Collection

/plans/{planId}

Fields

ownerId

status

createdAt

completedAt

route

budget

duration

weather

experienceScore

mood

transport

participants

summary

Route Object

Contains

routeId

stops

totalDistance

totalTime

estimatedCost

createdByAI

version

Stop Object

Contains

stopId

placeId

title

category

coordinates

arrivalTime

duration

estimatedPrice

status

memoryMoment

alternativePlaces

Albums Collection

/albums/{albumId}

Fields

ownerId

planId

title

coverPhoto

city

date

weather

mood

photos

timeline

shared

favorite

createdAt

updatedAt

Album Photos

Each photo

photoId

storagePath

caption

takenAt

coordinates (optional)

Never store image binaries in Firestore.

Only metadata.

Favorites

/favorites/{userId}

Contains

favoritePlaces

favoriteCollections

favoriteAlbums

favoriteRoutes

Use arrays only for small datasets.

Use subcollections when growth is expected.

Discover

/discover

Curated collections only.

Contains

featured

seasonal

trending

hiddenGems

roadTrips

updatedAt

Mostly read-only.

Managed by backend.

Notifications

/notifications/{notificationId}

Fields

userId

title

body

type

read

createdAt

deepLink

expiresAt

Feedback

/feedback/{feedbackId}

Fields

userId

type

message

appVersion

createdAt

status

Optional screenshot reference.

App Configuration

/app_config

Contains

minimumVersion

maintenanceMode

featureFlags

seasonalTheme

premiumConfig

No sensitive information.

Relationships

User

↓

Plans

↓

Albums

↓

Photos

User

↓

Favorites

↓

Places

↓

Collections

Plans reference users.

Albums reference plans.

Never duplicate ownership unnecessarily.

IDs

Use Firestore auto-generated IDs.

Never use sequential IDs.

Never expose internal IDs publicly.

Timestamps

Always use server timestamps.

Never trust client time.

Indexes

Composite indexes expected for

ownerId + createdAt

ownerId + favorite

status + createdAt

premium + createdAt (future)

Optimize queries before launch.

Pagination

Always paginate

Albums

Plans

Collections

Notifications

Never load unlimited documents.

Security

Every document has an owner.

Access verified through Firestore Security Rules.

No client-side authorization only.

Offline

Firestore offline cache enabled.

Conflict resolution

Last write wins

for MVP.

Future collaborative features may require merge strategies.

Migration Strategy

Every document includes

schemaVersion

Allows future migrations.

Never assume fixed schema forever.

Performance

Keep documents

< 1 MB

Prefer

< 100 KB

Load only required fields.

Lazy load large collections.

Future Collections

/friends

/groups

/shared_albums

/messages

/year_recaps

/adventure_maps

/events

Reserved for V2.

Delight Checklist

✓ Simple hierarchy.

✓ Scalable.

✓ Minimal reads.

✓ Easy to secure.

✓ Easy to extend.

DO NOT

Do NOT store API keys.

Do NOT store passwords.

Do NOT store images directly.

Do NOT duplicate large datasets.

Do NOT trust client timestamps.

Golden Rule

The database should disappear behind the product.

Developers should understand it immediately.

Users should never notice it.

Claude Instructions

Every new feature must define:

Which collection?

Which document?

Who owns it?

Who can read it?

Who can write it?

Never introduce new collections without updating this schema.

Maintain consistency at all times.

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